

SPACE INVADERS (1978)

Complete Restoration Guide for Public Placements

Cabinet • Controls • Power • Monitor • Reliability

Prepared for **Coin Op 4 Charity** / Arcade Angels

Preserving a foundational arcade classic for youth and community spaces

1. Overview & Goals

Space Invaders (Taito / Midway, 1978) is one of the most important arcade games ever made. For public and youth-center placements the priorities are reliability, clean presentation, and a smooth player experience. Many surviving machines use discrete-logic boards and black-and-white monitors with color overlays — this guide focuses on practical restoration steps that keep the machine playable and inviting.

2. Safety

- Unplug the cabinet before any internal work.
- CRT monitors store high voltage — discharge the anode before working on the chassis or neckboard.
- Original power supplies involve AC and higher voltages; treat every connection with care.
- If CRT work is outside your comfort zone, an LCD conversion is a practical option for high-use public sites.

3. Cabinet Exterior & Cosmetics

- Clean the entire cabinet thoroughly. Remove sticky residue and years of dust and hand prints.
- Inspect side art, front decals, bezel, and control-panel overlay. Replace or carefully restore heavily damaged graphics so the machine looks cared for.
- Many upright Space Invaders used a color overlay or plastic sheet over a black-and-white monitor. Clean or replace the overlay if it is yellowed, cracked, or missing sections.
- Marquee: clean the plastic and replace the fluorescent tube/starter if dim. A bright marquee makes the machine far more inviting.
- Coin door: clean, lubricate, confirm locks or convert to free-play for youth placements and document the setting.
- Level the cabinet so it sits solidly.

4. Control Panel, Buttons & Joystick / Buttons

Space Invaders typically uses left/right movement buttons (or a joystick on some conversions) and a fire button. Clean or replace sticky buttons. Inspect leaf switches or microswitches; clean contacts and adjust for positive, reliable actuation. A worn control-panel overlay can be replaced for both appearance and hygiene in public settings. Secure any lifting edges.

5. Power Supply & Edge Connector

Clean the power supply area thoroughly. Check and replace oxidized or burned fuse holders. Verify correct fuse values and stable voltages at the board. The edge connector is a frequent source of intermittent boots, resets, or missing power — clean the PCB fingers and the cabinet-side connector carefully. Many operators later convert to a modern switching supply for cleaner power and fewer connector issues. Either path is acceptable if voltages are solid under load.

6. Game Boards

Original Space Invaders hardware is discrete logic (and later CPU-based variants exist). Reseat boards firmly. Clean edge connectors. Look for obvious corrosion, broken traces, or burned components. If the machine has a history of random resets with power present, the edge connector and power path are the first places to check before assuming a deep board fault. Document any non-original boards or ROM sets so future volunteers know what is installed.

7. Monitor

CRT path

Original monitors were typically black-and-white with a color overlay. Common issues include weak brightness, focus problems, and chassis failures. Start with cleaning, checking monitor fuses, confirming isolation power, and reseating the neckboard. Only attempt deeper CRT work if experienced and following proper discharge procedures.

LCD conversion

An LCD conversion reduces heat, eliminates high-voltage hazards, and provides consistent picture quality. Kits and mounting solutions exist; some cabinet work is often required. Preserve the original CRT and chassis when possible — they remain useful as donor parts or for future authentic restoration.

8. Sound, Coin Door & Interior

- Confirm the speaker produces clean sound at reasonable volume. Replace if distorted or the cone is damaged.
- Clear dust from the interior, especially around the power supply and boards. Heat shortens component life.
- Confirm the rear interlock switch functions for safety.
- For free-play youth placements, set and document free-play or appropriate credit settings.

9. Final Testing Checklist Before Placement

- Full cold boot to attract mode with correct picture and sound
- Left/right movement and fire button register cleanly and repeatedly
- Complete several full games with no freezes or resets
- Monitor picture is focused and bright enough from normal player distance
- Color overlay (if present) is clean and properly aligned
- Cabinet exterior is clean, level, and free of sharp edges or loose parts
- Marquee is lit and readable
- Free-play or pricing configured correctly for the host site
- Keys, spare fuses, and basic notes bagged and labeled for the host organization

10. Notes for Host Sites

Provide a short one-page sheet: power on/off, free-play status, and who to contact. Recommend a weekly wipe-down of the control panel and glass. Ask the host to report new stickiness, picture changes, or resets early. Document any non-original parts (LCD, switching supply, overlay, etc.) for future volunteers.

This guide is intended for educational and nonprofit restoration use by Coin Op 4 Charity and partner organizations. The goal is a Space Invaders that stays reliable, safe, and fun for the people who play it.

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